



BRENT OZAR
UNLIMITED®

Faster, Cheaper Cloud Databases

Slides & scripts: 14days.brentozar.com/faster.zip

Download within 14 days. It'll disappear.

Deck 1 p1

Get the slides & scripts

14days.brentozar.com/faster.zip

Download it within 14 days – it'll disappear

Access problems: email Help@BrentOzar.com,
and I will respond to those after the class (not during)



Deck 1 p2

Asking questions during class

Raise your hand anytime! I love discussions!

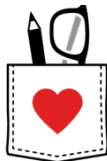
No private questions during bio breaks or lunch:
I need to pee too, and set up for the next module.

(Selfies are okay anytime too, and I have stickers.)

At the end of the day, we'll have an open Q&A.

No private questions there either: don't line up to
ambush me at the end of the day. Ask in the group.

Deck 1 p3



BRENT OZAR
UNLIMITED

99-05: dev, architect, DBA
05-08: DBA, VM, SAN admin
08-10: MCM, Quest Software
Since: consulting DBA

www.BrentOzar.com
Help@BrentOzar.com



Deck 1 p4



My job: 2-day SQL Critical Care®

Day 1, morning: rapidly assess a single SQL Server, database indexes, queries running against it, team

Day 1 afternoon & day 2 morning: write findings

Day 2 afternoon: deliver findings & training to get the team out of the emergency, quickly

Today: sharing query tuning techniques, experiences

Deck 1 p5



What you'll learn today

If your workload is small, medium, or large

How to pick the right VM type for your workload

How to size storage for performance, not size

How to assess wait stats in cloud VMs

How to back up directly to cloud-native file systems

Where other cloud database services make sense

Deck 1 p6



What you WON'T learn today

Which cloud provider is “better”
(I’m not into sports)

Which cloud is cheaper
(it doesn’t matter)

Which specific VM type to use on May 16, 2025
(that knowledge expires too quickly)

Deck 1 p7



All of this expires.

Today’s slides & demos are true as of summer 2025.

If you’re re-reading this deck in 2027 or beyond,
you should throw it away. Everything changes.

Deck 1 p8





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Assessing Your Cloud Performance Problems

Deck 1 p9

What we'll discuss

The 3 metrics to gather

How to gather those metrics quickly

Small/medium/large ranges for each metric

How to approach small/medium/large cloud problems



Deck 1 p10

3 performance metrics

3 performance metrics

1. How much data are we hosting?
**Total data size in GB/TB,
and number of databases.**
2. How fast are we going?
Batch Requests per second.
3. How hard are we working to achieve that speed?
**Wait time ratio over time
(like days of waits in a 1-day time span)**



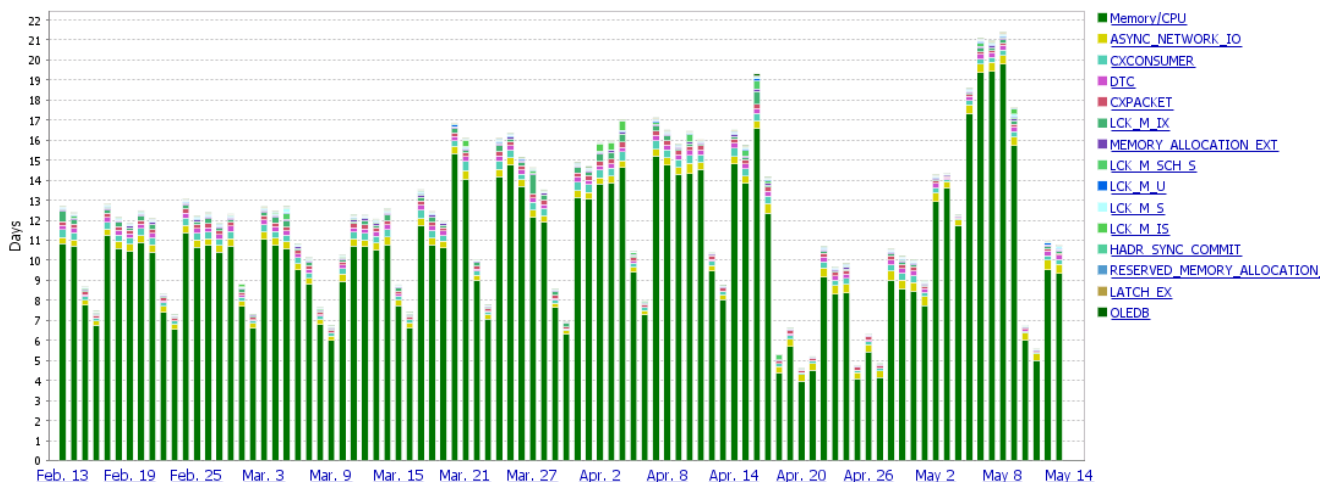
Explaining Wait Time Ratio

If the ratio is...	That means...
Zero	The server isn't waiting on anything.
0.25	In 1 second, queries wait 0.25 seconds on stuff. In 1 day, queries wait ¼ day (6 hours) on stuff.
0.5	In 1 day, queries wait ½ day (12 hours) on stuff.
1.0	In 1 day, queries wait 1 day on stuff.
2.0	In 1 day, queries wait 2 days on stuff. (We have multiple queries waiting all the time.)

Deck 1 p13



How Solarwinds displays it (great!)



Deck 1 p14



Generally speaking...

Whatever your top wait type is,
that's the bottleneck you want to focus on tuning.

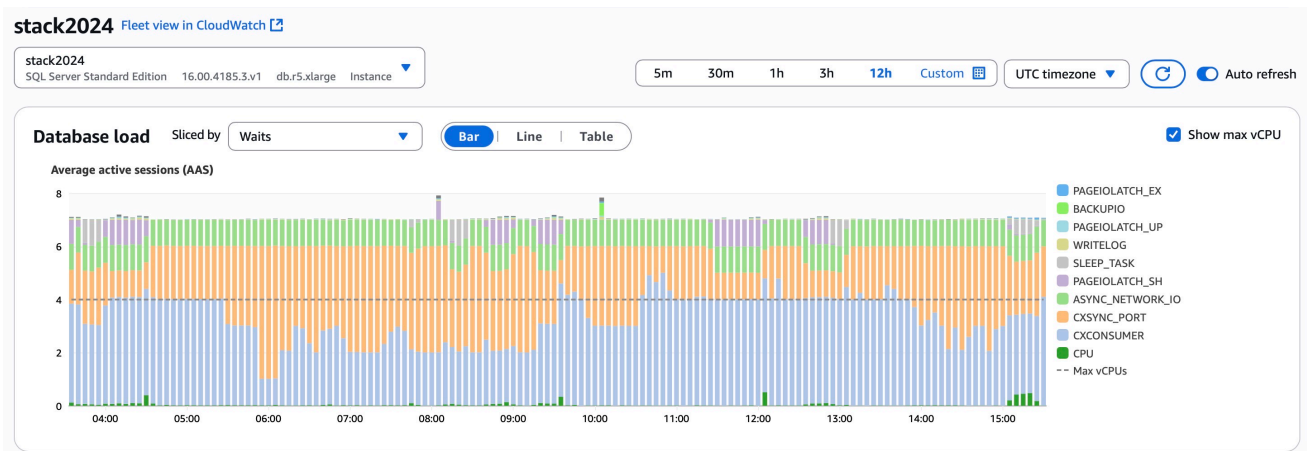
There are some exceptions:

- Poison waits like RESOURCE_SEMAPHORE and THREADPOOL
- Solarwinds' "Memory/CPU" just means queries are running, consuming CPU

Deck 1 p15



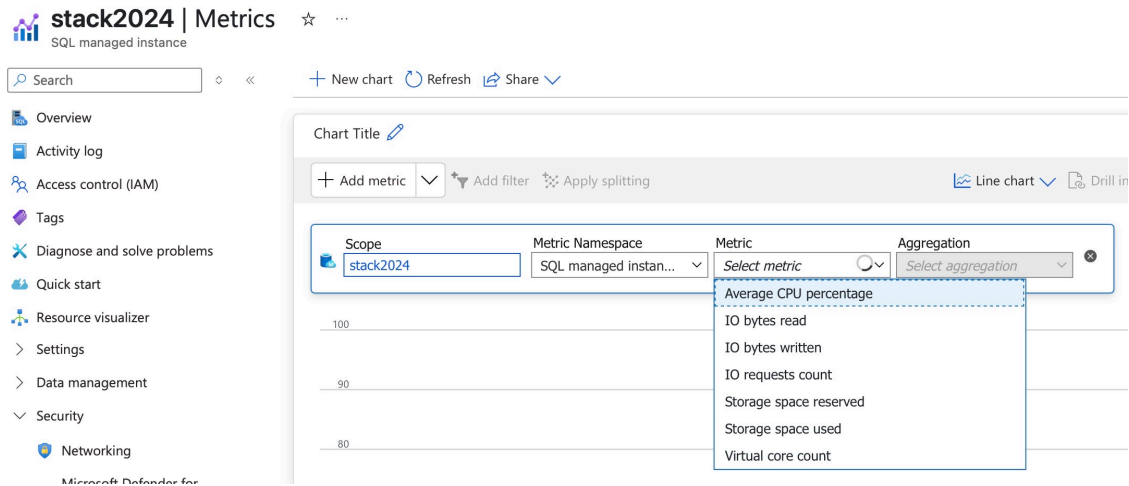
AWS PI displays it poorly



Deck 1 p16



Azure doesn't display waits at all!



Deck 1 p17



I don't blame the cloud vendors

Think about Microsoft SQL Server...

It's been 25 years now...

And still no wait stats visualization there either!

Deck 1 p18



3 ways to get wait stats data

1. Third party monitoring software
(Solarwinds, Quest, Idera, Redgate, etc.)
2. Build it yourself with your cloud provider's tools
(AWS Cloudwatch, Azure Managed Grafana, etc.)
3. `sp_BlitzFirst` from FirstResponderKit.org
(check out my free [How I Use the First Responder Kit](#) class for details on logging it to tables)

Deck 1 p19



Quick look: @SinceStartup = 1

```
1 sp_BlitzFirst @SinceStartup = 1
```

150 %

✓ No issues found

Results

Messages

	Pattern	Hours Sample	Thread Time (Hours)	wait_type	wait_category	Wait Time (Hours)	Avg ms Per Wait
1	WAIT STATS	218.0	2493.7	PAGEIOLATCH_SH	Buffer IO	1485.1	52.2
2	WAIT STATS	218.0	2493.7	CXCONSUMER	Parallelism	748.2	305.9
3	WAIT STATS	218.0	2493.7	CXSYNC_PORT	Other	214.7	52656.7
4	WAIT STATS	218.0	2493.7	ASYNC_NETWORK_IO	Network IO	15.4	137.5
5	WAIT STATS	218.0	2493.7	LCK_M_S	Lock	9.6	264472.7
6	WAIT STATS	218.0	2493.7	SOS_SCHEDULER_YIELD	CPU	6.3	1.7
7	WAIT STATS	218.0	2493.7	CXPACKET	Parallelism	3.5	4.7

Deck 1 p20



What the numbers mean

Hours Sample: uptime, how much history we have

Thread Time: how many hours of queries have been running during that time

- Includes all time from parallel queries
- Includes waiting time
- Is the same for all rows (just easy way to show)

Wait Time (Hours), Avg ms Per Wait: self-explanatory

Deck 1 p21



Sampling it over time

Run this in an SQL Agent job every 15 minutes, and it'll automatically keep 7 days of history:

```
EXEC sp_BlitzFirst
    @OutputDatabaseName = 'dbatools',
    @OutputSchemaName = 'dbo',
    @OutputTableName = 'BlitzFirst';
```

Can then visualize it, plus track median, max data for your peak workload times.

Deck 1 p22



Wait types can be cryptic

Wait_category: from Microsoft,
I don't use it, I disagree with some

Scroll to the far right for a URL
that links to SQLskills' wait library

wait_type	wait_category
PAGEIOLATCH_SH	Buffer IO
CXCONSUMER	Parallelism
CXSYNC_PORT	Other
ASYNC_NETWORK_IO	Network IO
LCK_M_S	Lock
SOS_SCHEDULER_YIELD	CPU

Covered extensively in my Mastering Server Tuning

Today we'll hit the most common cloud ones

Deck 1 p23



Size ranges for each metric

What's normal for SQL Server?

Microsoft doesn't publish metrics.

We do from SQL ConstantCare®:

<https://www.brentozar.com/sql-constantcare/>

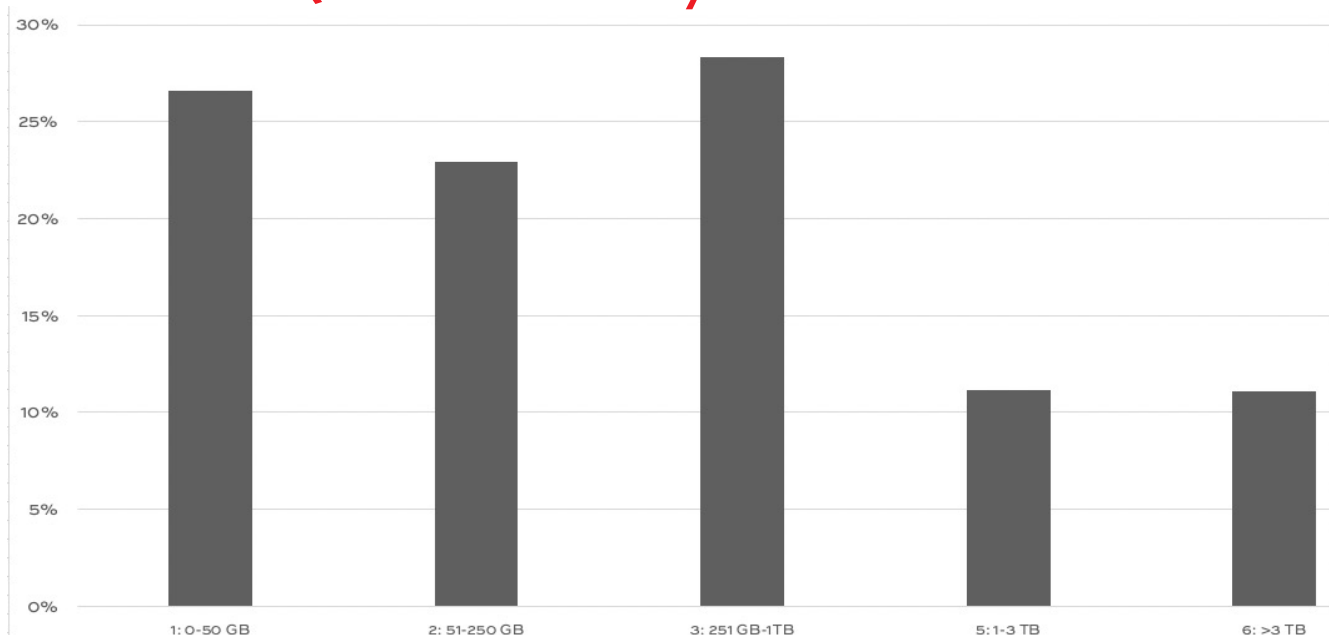
These metrics are biased:

- ~400 users, about 3,500 SQL Servers
- Only servers people care enough to monitor
- Only people who know about BrentOzar.com
- Tends to be smaller businesses, not enterprises

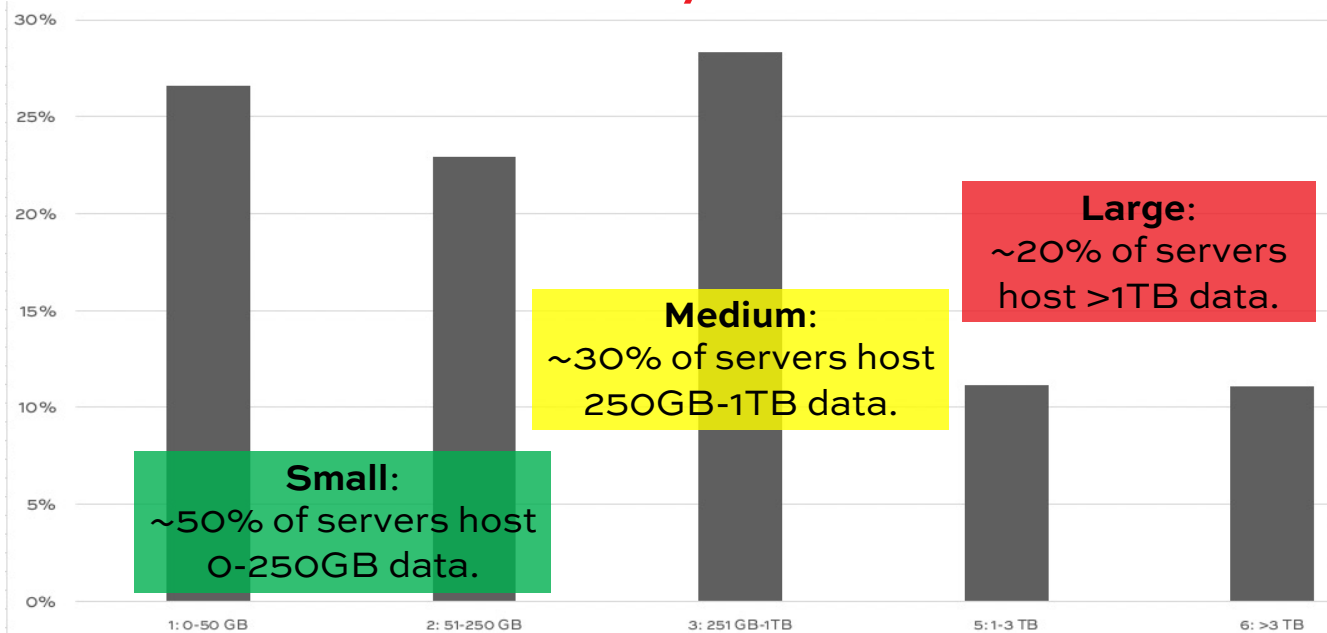
Deck 1 p25



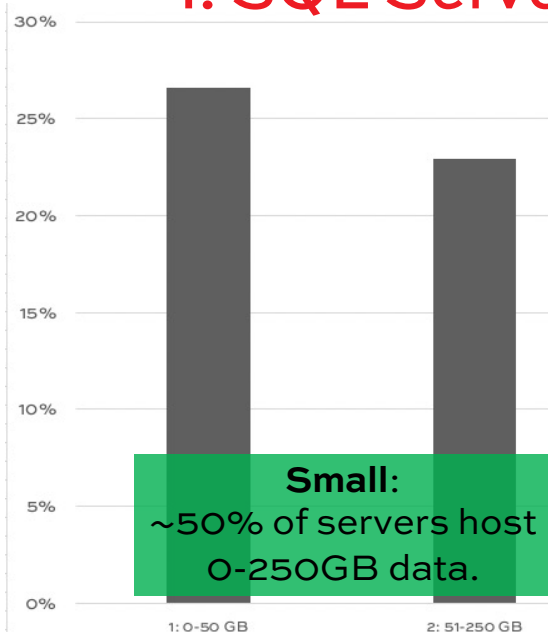
1. SQL Servers by total data size



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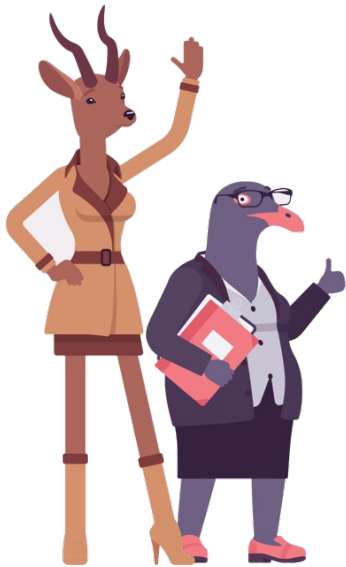
The cloud focused here first.

Doesn't take much power

Lots of these can coexist on the same hardware

Backup, restore, CHECKDB are easy and fast





That's why small businesses love the cloud.

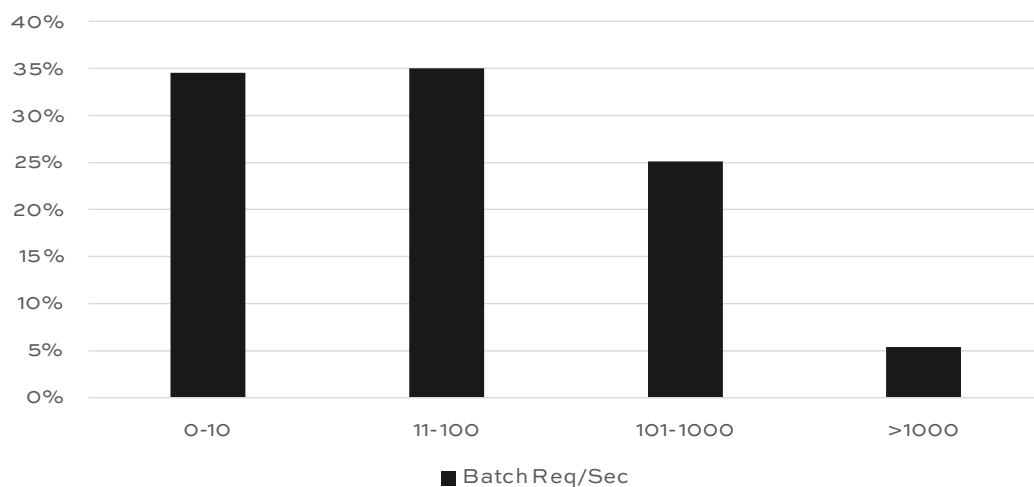
Their databases tend to be <1 TB.

Their databases tend to be standalone.

Deck 1 p29



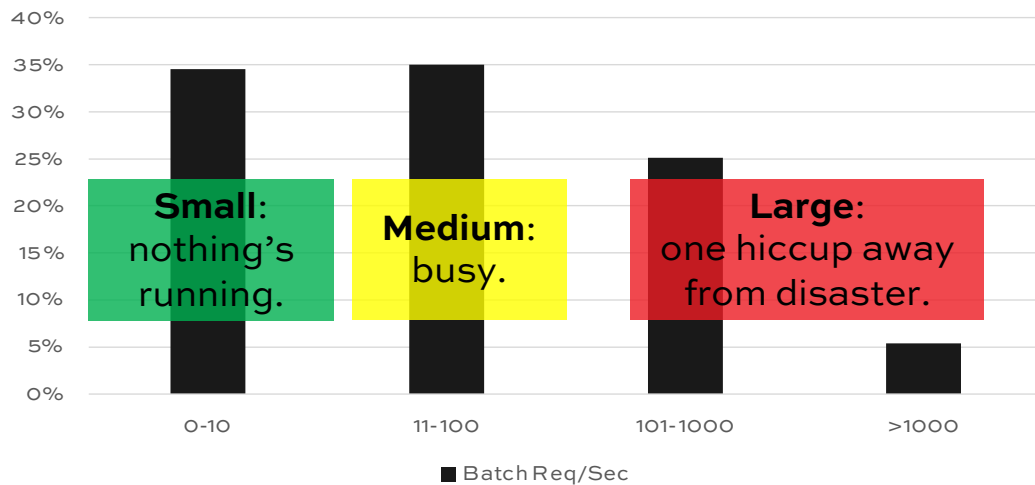
2. Batch Requests per Second



Deck 1 p30



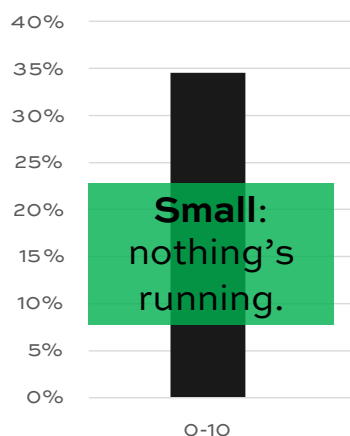
2. Batch Requests per Second



Deck 1 p31



2. Batch Requests per Second



The cloud focused here first.

Easy to find maintenance windows

Few transaction log issues

Easy to move from one hardware box to another

Doesn't need a full time DBA

Deck 1 p32



Again, great for small biz.

Small businesses don't have that many simultaneous users.

Their web sites tend to be hosted elsewhere, like SquareSpace, not on SQL Server.

They tend to not have active e-commerce sites.



Deck 1 p33



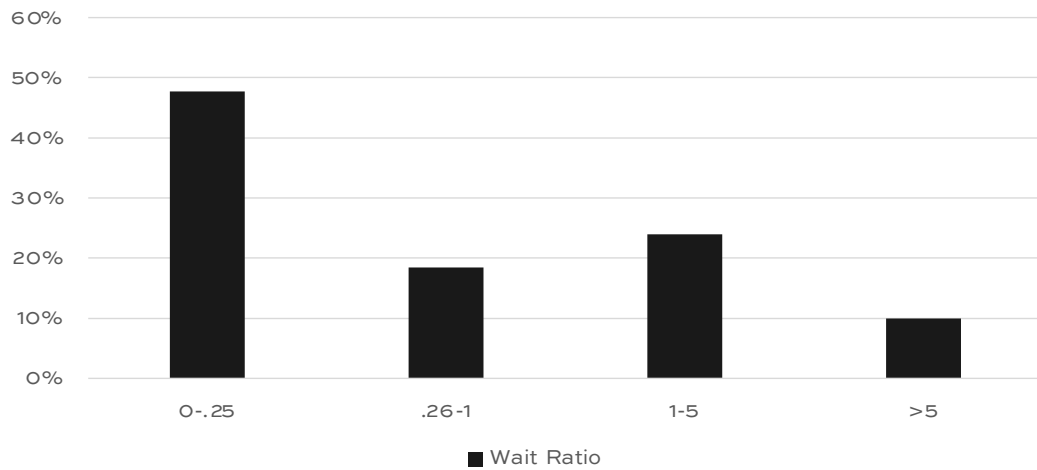
3. Wait Time Ratio

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0.5	In 1 day, queries wait 1/2 day (12 hours) on stuff.
1.0	In 1 day, queries wait 1 day on stuff.
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Deck 1 p34



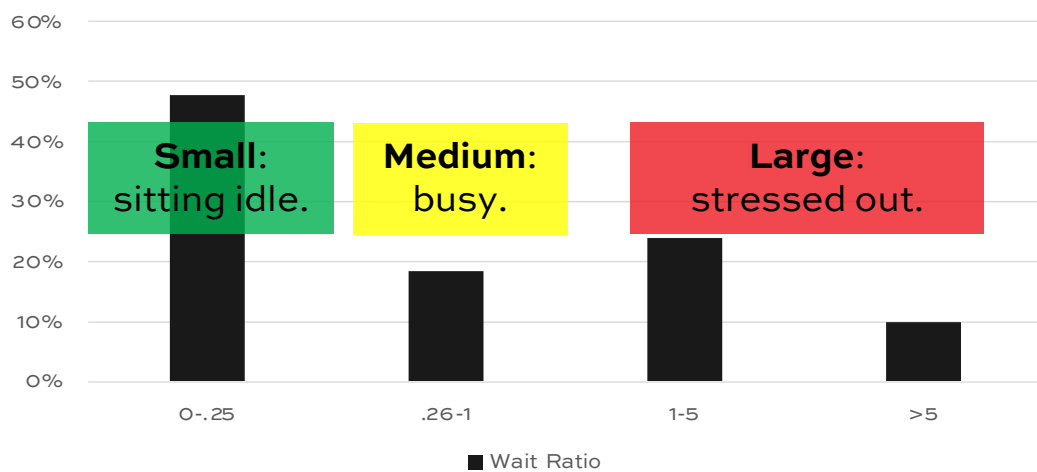
3. Wait Time Ratio



Deck 1 p35



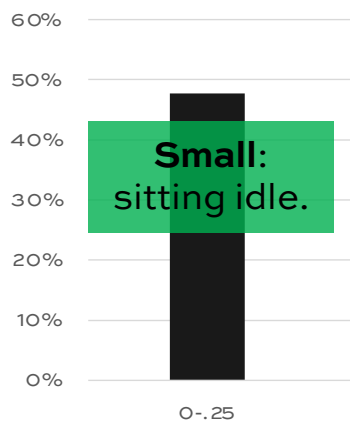
3. Wait Time Ratio



Deck 1 p36



3. Wait Time Ratio



The cloud focused here first.

Easy to find maintenance windows

Low hardware requirements

Profitable for MS to host

Doesn't need a full time DBA

Deck 1 p37



Ranking summary

	Small	Medium	Large
1. Total size, GB	0-250	250-1TB	>1TB
2. Batch req/sec	0-10	10-100	>100
3. Wait time ratio	0-0.25	0.25-1	>1
PaaS will work	Yes	With some work	Maybe not
IaaS VMs will work	Yes	With some work	Maybe not

Gather with your monitoring software or
`sp_BlitzFirst @SinceStartup = 1`

Deck 1 p38



Anybody could be green or red.



Deck 1 p39

You're probably a mix of colors.

	Small	Medium	Large	You
1. Total size, GB	0-250	250-1TB	>1TB	0-250
2. Batch req/sec	0-10	10-100	>100	10-100
3. Wait time ratio	0-0.25	0.25-1	>1	>1
PaaS will work	Yes	With some work	Maybe not	?
IaaS VMs will work	Yes	With some work	Maybe not	?

Deck 1 p40

You're probably a mix of colors.

	Small	Medium	Large	You
1. Total size, GB	0-250	250-1TB	>1TB	>1TB
2. Batch req/sec	0-10	10-100	>100	0-10
3. Wait time ratio	0-0.25	0.25-1	>1	0-0.25
PaaS will work	Yes	With some work	Maybe not	?
IaaS VMs will work	Yes	With some work	Maybe not	?

Deck 1 p41



Understanding the colors

The more red you have,
the more likely you are to have server-level problems,
and you're probably gonna need more horsepower.

The more green you have,
the more likely it is that user complaints
are really about query-level issues,
and you don't wanna throw hardware at it.

Deck 1 p42



How to approach small/medium/large problems

We have 3 dials to handle problems.



Table & Index Design



Query Design



Hardware & Licensing



I'll explain what \$\$\$\$ looks like.

For each dial, I'll talk about clients.

I'm by no means suggesting you need invest that much – but I need you to understand what an extreme investment might look like.

For every one, I have clients doing this exact thing – feel free to ask questions about 'em.

This will help you understand where you're at.



Deck 1 p45



What does \$\$\$\$ look like here?



Table & Index Design



Query Design



Hardware & Licensing

Deck 1 p46



What \$\$\$\$ looks like

Constant, careful analysis of how users write queries, and designing tables & indexes to best support those queries

Both normalization and denormalization where appropriate, like tables designed specifically for reporting or searching

A core set of indexes for basic client functionality, and then custom per-client indexes for their use cases

At the \$\$\$\$ volume level, it's 1-2 people doing this full time on an app, coaching the other staff on where to find the right tables for each query

Deck 1 p47



Example client: midsize company

100% in the cloud (Azure Managed Instances)

About 50 developers overall, split across states

Building internal apps & reporting tools for sales, warehousing, budgeting

Team of 3 full time development DBAs who:

- Review & approve all new tables
(and sometimes steer things over to NoSQL instead)
- Are responsible for all nonclustered indexes

Deck 1 p48



Next up: the query design dial



Table & Index Design



Query Design



Hardware & Licensing

Deck 1 p49



What \$\$\$\$ looks like

Constant, careful analysis of how users write queries, and designing tables & indexes to best support those queries

Both normalization and denormalization where appropriate, like tables designed specifically for reporting or searching

A core set of indexes for basic client functionality, and then custom per-client indexes for their use cases

At the \$\$\$\$ volume level, it's 1-2 people doing this full time on an app, coaching the other staff on where to find the right tables for each query

Deck 1 p50



Example client: e-commerce .com

Mix of cloud and on-premises

About 100 developers overall, split across states

2 full time development DBAs who:

- Focus on query performance monitoring tools
- When new resource-intensive queries pop up, they rewrite the queries for the developers, hand them over for source control & deployment
- Conduct monthly lunch & learn sessions with developers to coach them into writing better queries

Deck 1 p51



Finally, the hardware & licensing dial



Table & Index Design



Query Design



Hardware & Licensing

Deck 1 p52



Hardware/licensing dial: \$\$\$\$

Region	Pricing Unit	Cost	Reserved	Visible																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																													</	
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u-24tb1.112xlarge: 448 cores, 24TB RAM, \$1.3M/year

Available in AWS since 2019: <https://aws.amazon.com/blogs/aws/ec2-high-memory-update-new-18-tb-and-24-tb-instances/>

Deck 1 p53



Azure Managed Instances

Premium-series, memory-optimized – Zone Redundant

Premium-series logical CPUs are based on the latest Intel(R) Xeon Scalable 2.8 GHz processor (Ice Lake), 1 vCore = 1 hyper thread. This memory-optimized version provides nearly twice the memory per vCore and is great for database workloads that require even better memory performance than the standard next generation offering. Zone redundancy provides enhanced availability by automatically replicating data across the availability zones within a region. For the business critical tier, this option incurs an additional cost.

vCORE	Memory (GB)	Included Storage	Pay as you go	1 year reserved capacity	3 year reserved capacity
80	870.4	First 32 GB/month	\$67,977.60/month	\$51,846.352/month ~24% savings	\$42,633.168/month ~37% savings
96	870.4	First 32 GB/month	\$81,573.12/month	\$62,215.623/month ~24% savings	\$51,159.802/month ~37% savings
128	870.4	First 32 GB/month	\$108,764.160/month	\$82,954.164/month ~24% savings	\$68,213.069/month ~37% savings

\$1M/year with a reservation.

(This seems like less hardware for the buck than AWS, but remember that the AWS number doesn't include SQL Server licensing.)

Deck 1 p54



That's not even the full number

SQL Server licensing

Storage pricing

Replicas for high availability & disaster recovery

For hardware/licensing, \$\$\$\$ means
multiple millions of dollars per year
for one critical set of databases.

Deck 1 p55



Example client doing \$\$\$\$

Software-as-a-service (SaaS) vendor

App lets lawyers write queries with a drag & drop tool:
no way to tune queries or index for them in advance

Data being loaded & queried simultaneously

Every activity & change is logged to tables forever,
and the history is constantly analyzed too

Deck 1 p56



Another example: e-commerce co

Normally low-volume, but very heavy volume during:

- November-December (Black Friday, holidays)
- Annual sales (say March, June)

Availability Groups year-round

Near peaks: add new \$\$\$\$ replicas, fail over to them

After peaks: fail back over to cheap replicas,
shut down & delete the \$\$\$\$ servers

Deck 1 p57



So... what does ONE \$ look like?



Table & Index Design



Query Design



Hardware & Licensing

Deck 1 p58



Table & index design, one \$:

No one on staff has been to a formal SQL Server training class in the last 5 years

No one is personally accountable for monthly index review, tuning

Tables aren't archived, revisited, modernized

“We all kinda work on performance” = no one does

Deck 1 p59



Query design, one \$:

No one on staff has been to a formal SQL Server training class in the last 5 years

Queries are written with ORMs, never hand tuned

No one is personally accountable for knowing the top 10 most resource-intensive queries and doing anything about it

“We all kinda work on performance” = no one does

Deck 1 p60



Hardware & licensing, one \$:



Lenovo ThinkBook 16 G7 16" FHD+ Business Laptop Computer, Intel 16-Core Ultra 7 155H (Beat i7-13705H), Arc Graphics, 64GB DDR5 RAM, 2TB PCIe SSD, WiFi 6E, Backlit Keyboard, Windows 11 Pro, AZ-XUT

[Visit the Lenovo Store](#)
4.5 ★★★★★ (23) | Search this page
300+ bought in past month

\$1,149⁰⁰
Or \$52.63 /mo (36 mo). Select from 1 plan
prime One-Day
FREE Returns

[Unlock a \\$150 Amazon Gift Card](#) upon approval for Prime Visa.

May be available at a lower price from [other sellers](#), potentially without free Prime shipping.

Capacity: 64GB DDR5 RAM, 2TB PCIe SSD

32GB DDR5 RAM, 1TB...	64GB DDR5 RAM, 1TB...	64GB DDR5 RAM, 2TB...	64GB DDR5 RAM, 4TB...
\$999.00 FREE Delivery Tomorrow	\$1,099.00 FREE Delivery Tomorrow	\$1,149.00 FREE Delivery Tomorrow	\$1,299.00 FREE Delivery Tomorrow

Deck 1 p61



So, where are you today?



Table & Index Design



Query Design



Hardware & Licensing

Deck 1 p62



Ranking summary

	Small	Medium	Large
1. Total size, GB	0-250	250-1TB	>1TB
2. Batch req/sec	0-10	10-100	>100
3. Wait time ratio	0-0.25	0.25-1	>1
PaaS will work	Yes	With some work	Maybe not
IaaS VMs will work	Yes	With some work	Maybe not

Gather with your monitoring software or
`sp_BlitzFirst @SinceStartup = 1`

Deck 1 p63



Now, time to reflect.

3 performance metrics

1. How much data are we hosting?
**Total data size in GB/TB,
and number of databases.**
2. How fast are we going?
Batch Requests per second.
3. How hard are we working to achieve that speed?
**Wait time ratio over time
(like days of waits in a 1-day time span)**

Deck 1 p65



3 ways to track those metrics

1. Third party monitoring software
(Solarwinds, Quest, Idera, Redgate, etc.)
2. Build it yourself with your cloud provider's tools
(AWS Cloudwatch, Azure Managed Grafana, etc.)
3. sp_BlitzFirst from FirstResponderKit.org
(check out my free How I Use the First Responder
Kit class for details on logging it to tables)
- ~~4. Cloud vendor portal defaults~~

Deck 1 p66



3 categories for those metrics

	Small	Medium	Large
1. Total size, GB	0-250	250-1TB	>1TB
2. Batch req/sec	0-10	10-100	>100
3. Wait time ratio	0-0.25	0.25-1	>1

But your own databases might be a combination of different colors for different metrics.

You just wanna know roughly what group you're in.

Deck 1 p67



We have 3 dials to handle problems.



Table & Index Design



Query Design



Hardware & Licensing

Deck 1 p68

